Laura Marks and Stephen Makonin: Streaming video is overheating the planet

Opinion: We can no longer ignore the carbon footprint resulting from the glut of streaming video

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Joe Exotic was quite the character in the Netflix blockbuster Tiger King, apparently generating a lot of electricity as North Americans streamed the popular show.

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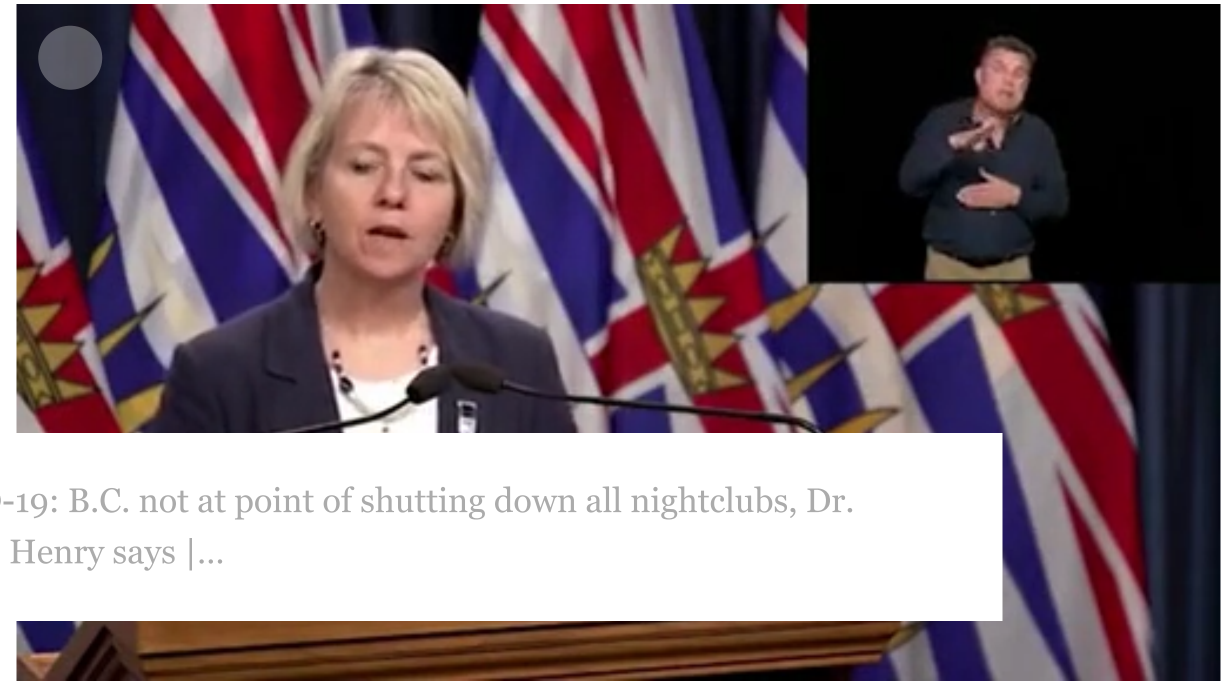
In the early days of the COVID-19 pandemic, people self-isolating around the world took comfort by watching online movies and TV shows.

Over 10 days in March, 34 million locked-down Americans streamed Tiger King on Netflix. Those streams, we calculated, consumed half a terawatt hour — equal to the annual electrical consumption of Rwanda in 2016. Put another way, the national Tiger King binge was equivalent to the emissions of 75,834 passenger cars for one year, given the U.S.'s reliance on fossil fuels for electricity.



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We can no longer ignore the carbon footprint resulting from the glut of streaming video, from sources including video-on-demand services, YouTube and other 'Tubes, pornography, websites, online games and now Zoom meetings. Information and communications technology is estimated to generate about three per cent of the greenhouse gases (GHG) that cause global warming. (For comparison, the airline industry generates 1.9 per cent of GHG.)

The French think tank The Shift Project estimates that streaming video alone contributes one per cent of global GHG. The International Energy Association, a consortium that it includes many fossil fuel companies, recently attacked The Shift Project report, drawing very selectively on engineering research to minimize the problem.

Although 90 per cent of B.C.'s energy comes from hydroelectric power, the streaming media platforms that serve B.C. consumers are located elsewhere, often in the U.S., and largely powered by fossil fuels.

It's still common to think of online media as "virtual." But we need to understand the physical infrastructure behind this seemingly innocuous presence if we are to take seriously our commitment to reducing our environmental impact. Streaming video is transmitted between data centres, along wired and wireless networks, to digital televisions, computers and phones. At each point, the electricity they use generates significant amounts of CO_2 — significantly more if we factor in the production and disposal of smartphones and other devices.

Companies that produce Internet hardware and software like Cisco and Dell, and media corporations like Netflix, Amazon, and Alphabet, employ thousands of engineers and programmers tasked with making their services more energy efficient.

But gains in efficiency are hard won, as Moore's Law — the observation that the number of transistors in a dense integrated circuit doubles about every two years — and even Koomey's Law, on the increasing electrical efficiency of computation, do not apply to the physical world of Internet infrastructure hardware.

What's important to us is that gains in efficiency are outpaced by rising demand for streaming video, driven by these same companies. They want us to stream more video, on more devices, at higher resolution, requiring higher bandwidth. Increased efficiency translates to more streaming, not less: as the network corporation Cisco says, "Broadband-speed improvements result in increased consumption and use of high-bandwidth content and applications." This is known as the Jevons paradox: More efficient technologies often encourage greater use of a resource, reducing or eliminating savings. If you build it, they will come.

As well as the sheer volume of streaming, the main culprit of our massive streaming carbon footprint is consumers' desire for ever higher resolution — HD, 4K, even 8K video. It's worth asking ourselves why we so long for detailed moving pictures.

The Shift Project advocates "digital sobriety": Consuming less streaming video, more consciously, at lower resolution. If this kind of sobriety doesn't appeal, we encourage you to lobby your governments to speed the conversion to renewable energy.

Media scholar Laura Marks and IT engineer Stephen Makonin are professors at Simon Fraser University. Their research project, "Tackling the Carbon Footprint of Streaming Media," translates engineering and media industry data for lay audiences, to raise awareness and influence policy. Marks is the founder of the First Annual Small File Media Festival, streaming in small bandwidth Aug. 10-20.

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